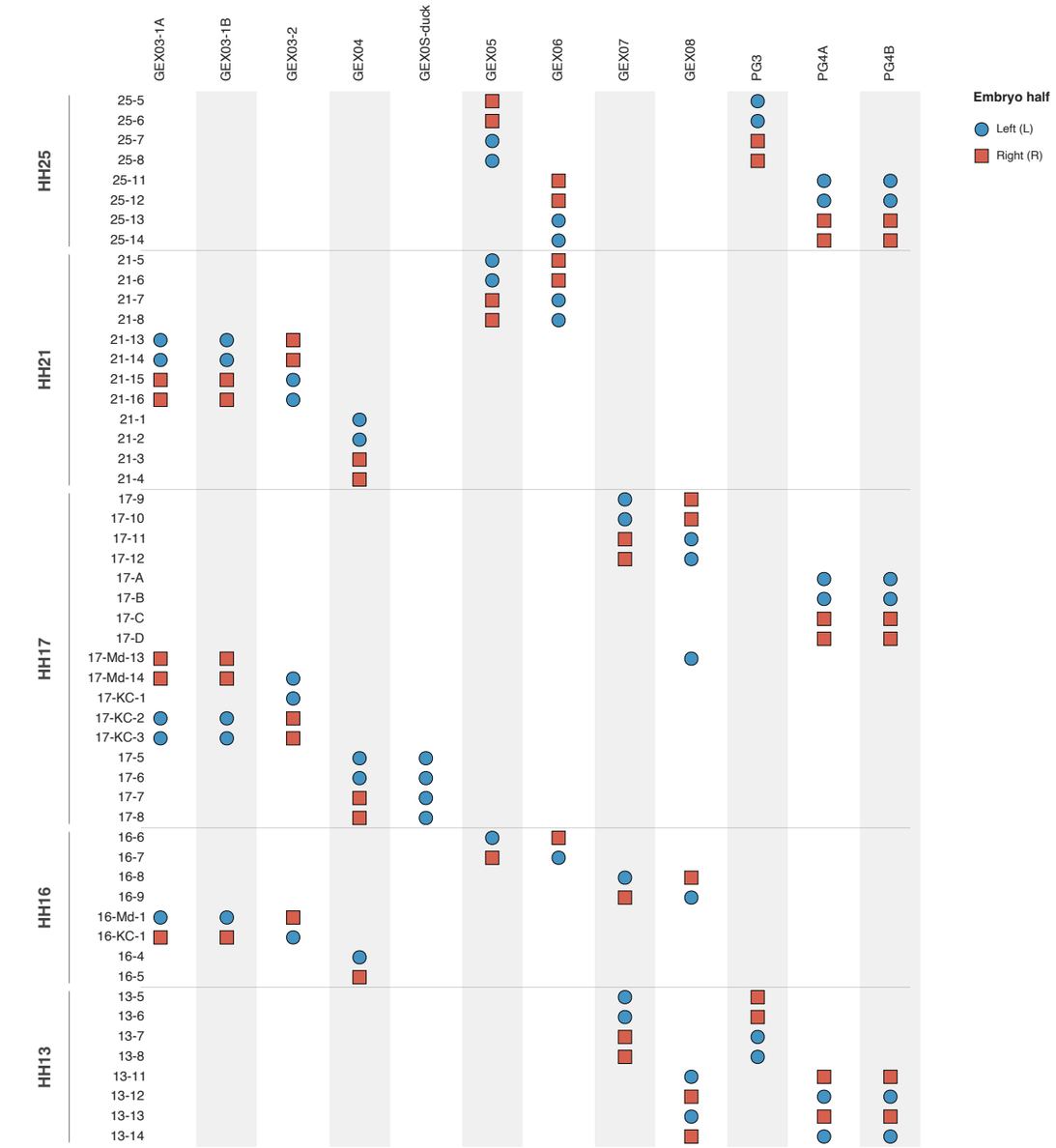
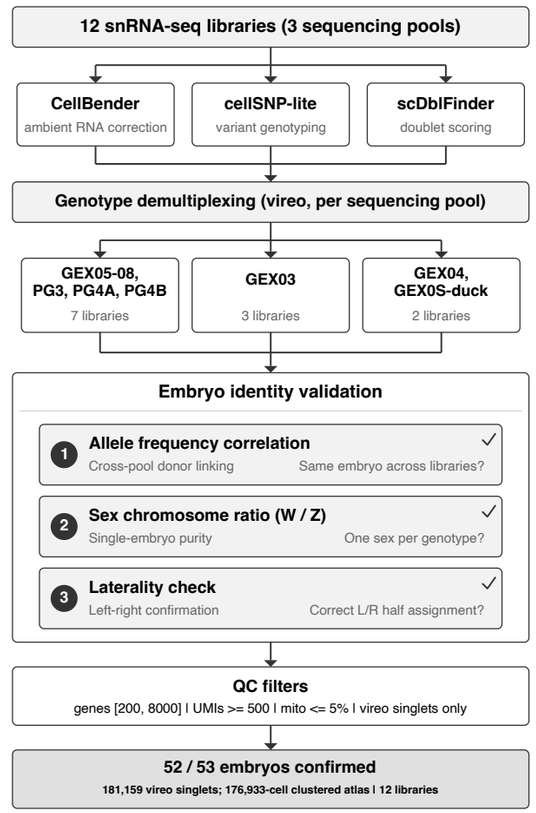
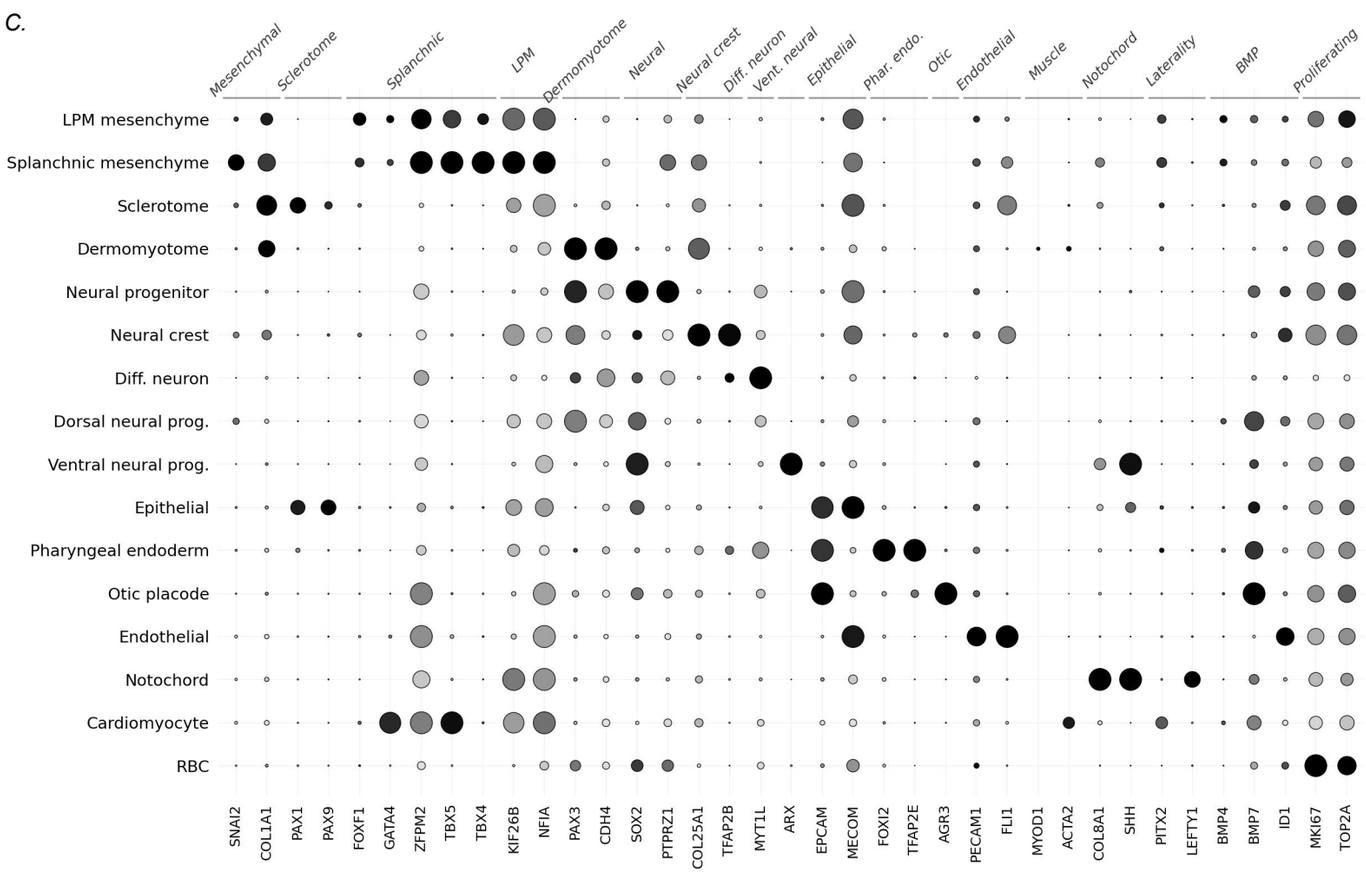
A.


GEX03-1A
GEX03-1B
GEX03-2
GEX04
GEX0S-duck
GEX05
GEX06
GEX07
GEX08
PG3
PG4A
PG4B
HH25
25-5
25-6
25-7
25-8
25-11
25-12
25-13
25-14
HH21
21-5
21-6
21-7
21-8
21-13
21-14
21-15
21-16
21-1
21-2
21-3
21-4
HH17
17-9
17-10
17-11
17-12
17-A
17-B
17-C
17-D
17-Md-13
17-Md-14
17-KC-1
17-KC-2
17-KC-3
17-5
17-6
17-7
17-8
HH16
16-6
16-7
16-8
16-9
16-Md-1
16-KC-1
16-4
16-5
HH13
13-5
13-6
13-7
13-8
13-11
13-12
13-13
13-14
Embryo half
Left (L)
Right (R)

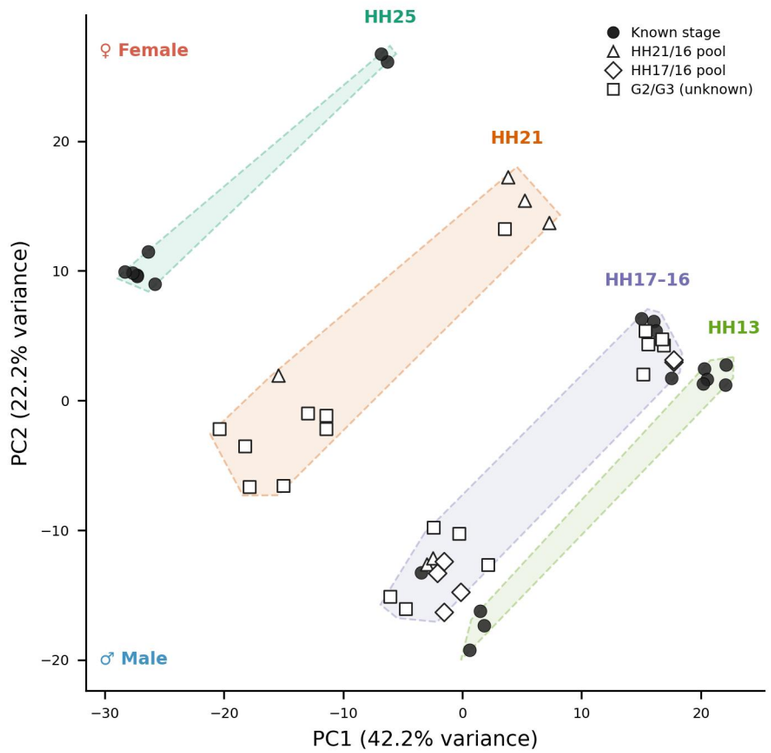
B.


12 snRNA-seq libraries (3 sequencing pools)
CellBender
ambient RNA correction
cellSNP-lite
variant genotyping
scDblFinder
doublet scoring
Genotype demultiplexing (vireo, per sequencing pool)
GEX05-08, PG3, PG4A, PG4B
7 libraries
GEX03
3 libraries
GEX04, GEX0S-duck
2 libraries
Embryo identity validation
1 Allele frequency correlation
Cross-pool donor linking
Same embryo across libraries?
2 Sex chromosome ratio (W / Z)
Single-embryo purity
One sex per genotype?
3 Laterality check
Left-right confirmation
Correct L/R half assignment?
QC filters
genes [200, 8000] | UMIs >= 500 | mito <= 5% | vireo singlets only
52 / 53 embryos confirmed
181,159 vireo singlets; 176,933-cell clustered atlas | 12 libraries

C.


Mesenchymal
Sclerotome
Splanchnic
LPM
Dermomyotome
Neural
Neural crest
Diff. neuron
Vent. neural
Epithelial
Phar. endo.
Otic
Endothelial
Muscle
Notochord
Laterality
BMP
Proliferating
LPM mesenchyme
Splanchnic mesenchyme
Sclerotome
Dermomyotome
Neural progenitor
Neural crest
Diff. neuron
Dorsal neural prog.
Ventral neural prog.
Epithelial
Pharyngeal endoderm
Otic placode
Endothelial
Notochord
Cardiomyocyte
RBC
SNAI2
COL1A1
PAX1
PAX9
FOXF1
GATA4
ZFPM2
TBX5
TBX4
KIF26B
NFIA
PAX3
CDH4
SOX2
PTPRZ1
COL25A1
TFAP2B
MYT1L
ARX
EPCAM
MECOM
FOXI2
TFAP2E
AGR3
PECAM1
FLI1
MYOD1
ACTA2
COL8A1
SHH
PITX2
LEFTY1
BMP4
BMP7
ID1
MKI67
TOP2A
Fraction expressing
10%
30%
50%
Mean expression in group
1.0
0.5
0.0

D.


♀ Female
♂ Male
HH25
HH21
HH17–16
HH13
Known stage
HH21/16 pool
HH17/16 pool
G2/G3 (unknown)
PC1 (42.2% variance)
PC2 (22.2% variance)
20
10
0
−10
−20
−30
−20
−10
0
10
20